康普顿散射.

$$\begin{cases} p^2 = p_\gamma^2 + p_{\gamma'}^2 - 2p_\gamma p_{\gamma'}\cos\theta & ① \\ E_\gamma - E_{\gamma'} = E - m_0c^2 & ② \end{cases}$$

① $\Rightarrow$ $\dfrac{E^2 - m_0^2c^4}{c^2} = \dfrac{E_\gamma^2}{c^2} + \dfrac{E_{\gamma'}^2}{c^2} - 2\dfrac{E_\gamma E_{\gamma'}}{c^2}\cos\theta$

即

$$E^2 - m_0^2c^4 = E_\gamma^2 + E_{\gamma'}^2 - 2E_\gamma E_{\gamma'}\cos\theta.$$

又 ②: $E = E_\gamma - E_{\gamma'} + m_0c^2$, 代入①.

$$(E_\gamma - E_{\gamma'})^2 + 2(E_\gamma - E_{\gamma'})m_0c^2 = E_\gamma^2 + E_{\gamma'}^2 - 2E_\gamma E_{\gamma'}\cos\theta.$$

$\Leftrightarrow$ $$-2E_\gamma E_{\gamma'} + 2(E_\gamma - E_{\gamma'})m_0c^2 = -2E_\gamma E_{\gamma'}\cos\theta.$$

$\Leftrightarrow$ $$(E_\gamma - E_{\gamma'})m_0c^2 = (1-\cos\theta)E_\gamma E_{\gamma'}$$

$$E_{\gamma'} = \frac{E_\gamma m_0c^2}{(1-\cos\theta)E_\gamma + m_0c^2}$$